

- **Evaluation:**
 - **Root Mean Squared Error (RMSE):** The model achieved a low RMSE, indicating its predictions are, on average, very close to the actual prices.
 - **R-squared (R²):** An R² value of **0.92** was achieved, meaning the model explains 92% of the variance in the price data.

4.3.2 Offer Generation Logic

The offer system is rule-based, triggered by the pricing model's inputs:

- IF move_day_of_week IN (Monday, Tuesday, Wednesday) THEN apply 10% weekday_discount.
- IF move_hour BETWEEN 11 AND 16 THEN apply 5% off_peak_discount.
- IF booking is made > 14 days in advance THEN apply 7% early_bird_discount.

4.4 Module 4: Conversational AI Chatbot

4.4.1 NLP Engine and Intent Recognition

The open-source framework **Rasa** was used to build the chatbot.

- **NLU Training Data (nlu.yml):** We defined intents and provided example user utterances.

```
version: "3.1"
nlu:
  - intent: greet
    examples: |
      - hey
      - hello
  - intent: track_booking
    examples: |
      - Where is my stuff?
```